

In-Text Exercises from Ch.3: Tangent Vectors, Lee

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Disclaimer: The proofs given are completed by myself. Thus, they are also reviewed by myself to ensure no logic flaws and incorrect arguments/deductions/conclusions. If you find any error in my proofs, please do not hesitate to contact me so we can discuss any potential errors and I can update the document as needed.

In-Text Exercises

Exercise 0.1. (Exercise 3.5)

Prove Lemma 3.4.

Proof. We state lemma:

Lemma 0.1. (Properties of Tangent Vectors on Manifolds)

Suppose M is a smooth manifold with or without boundary, $p \in M$, $v \in T_p M$, and $f, g \in C^\infty(M)$.

- (a) If f is a constant function, then $vf = 0$.
- (b) If $f(p) = g(p) = 0$, then $v(fg) = 0$.

We now prove this.

- a) Let $p \in M$, $v \in T_p M$, and $f \in C^\infty(M)$ be a constant function. It suffices to prove the statement for the constant function $f_1(x) \equiv 1$ since any other constant function $f(x) \equiv c$ can be written as cf_1 . By the product rule,

$$vf_1 = v(f_1 f_1) = f_1(p)vf_1 + f_1(p)vf_1 = 2vf_1$$

which implies that $vf_1 = 0$. Hence, by linearity, $vf = v(cf_1) = cvf_1 = 0$ as desired.

- b) Let $f, g \in C^\infty(M)$ such that $f(p) = g(p) = 0$. Then

$$v(fg) = f(p)vg + g(p)vf = 0 + 0 = 0.$$

□

Exercise 0.2. (Exercise 3.7)

Prove Proposition 3.6.

Proof. We state the proposition:

Proposition 0.1. (Properties of Differentials)

Let M, N , and P be smooth manifolds with or without boundary, let $F: M \rightarrow N$ and $G: N \rightarrow P$ be smooth maps, and let $p \in M$.

- (a) $dF_p: T_p M \rightarrow T_{F(p)} N$ is linear.
- (b) $d(G \circ F)_p = dG_{F(p)} \circ dF_p: T_p M \rightarrow T_{G \circ F(p)} P$.
- (c) $d(\text{Id}_M)_p = \text{Id}_{T_p M}: T_p M \rightarrow T_p M$.

- (d) If F is a diffeomorphism, then $dF_p: T_pM \rightarrow T_{F(p)}N$ is an isomorphism, and $(dF_p)^{-1} = d(F^{-1})_{F(p)}$.

We now proceed to the proof.

- a) Let $v, w \in T_pM$, $f \in C^\infty(N)$, and $a, b \in \mathbb{R}$. Then

$$\begin{aligned} dF_p(av + bw)(f) &= (av + bw)(f \circ F) \\ &= av(f \circ F) + bw(f \circ F) \\ &= a(dF_p(v)(f)) + b(dF_p(w)(f)). \end{aligned}$$

Hence, $dF_p: T_pM \rightarrow T_{F(p)}N$ is linear over $\mathbb{R}$.

- b) Let $v \in T_pM$ and $h \in C^\infty(P)$. Then

$$d(G \circ F)_p(v)(h) = v(h \circ (G \circ F)) \in \mathbb{R}.$$

Conversely,

$$\begin{aligned} (dG_{F(p)} \circ dF_p)(v)(h) &= (dG_{F(p)}(dF_p(v)))(h) \\ &= dF_p(v)(h \circ G) \\ &= v((h \circ G) \circ F) = v(h \circ (G \circ F)) \in \mathbb{R}. \end{aligned}$$

Therefore, for all $h \in C^\infty(P)$,

$$d(G \circ F)_p(v)(h) = (dG_{F(p)} \circ dF_p)(v)(h)$$

which implies that

$$d(G \circ F)_p(v) = (dG_{F(p)} \circ dF_p)(v)$$

for all $v \in T_pM$. Hence,

$$d(G \circ F)_p = dG_{F(p)} \circ dF_p: T_pM \rightarrow T_{G \circ F(p)}P,$$

as desired. One can view this scenario by the following commutative diagrams.

$$\begin{array}{ccc} M & & T_pM \\ F \downarrow & \searrow G \circ F & \searrow d(G \circ F)_p \\ N & \xrightarrow{G} & P \end{array} \quad \begin{array}{ccc} T_pM & & T_{G \circ F(p)}P \\ dF_p \downarrow & \searrow d(G \circ F)_p & \searrow dG_{F(p)} \\ T_{F(p)}N & \xrightarrow{dG_{F(p)}} & T_{G \circ F(p)}P \end{array}$$

- c) Given $v \in T_pM$ and $f \in C^\infty(M)$,

$$d(\text{Id}_M)_p(v)(f) = v(f \circ \text{Id}_M) = v(f) \in \mathbb{R}.$$

Conversely,

$$\text{Id}_{T_pM}(v)(f) = v(f).$$

Hence $d(\text{Id}_M)_p(v)(f) = \text{Id}_{T_pM}(v)(f)$ for all $f \in C^\infty(M)$. Thus, $d(\text{Id}_M)_p(v) = \text{Id}_{T_pM}(v)$ for all $v \in T_pM$. Therefore,

$$d(\text{Id}_M)_p = \text{Id}_{T_pM}: T_pM \rightarrow T_pM.$$

- d) Assume that $F: M \rightarrow N$ is a diffeomorphism. Thus, $F^{-1}: N \rightarrow M$ exists and is smooth. In particular,

$$F^{-1} \circ F = \text{Id}_M \quad \text{and} \quad F \circ F^{-1} = \text{Id}_N .$$

Considering the differentials of these smooth maps at $p \in M$ and applying (b) and (c), we obtain

$$d(F^{-1} \circ F)_p = d(F^{-1})_{F(p)} \circ dF_p = d\text{Id}_M = \text{Id}_{T_p M}$$

and

$$d(F \circ F^{-1})_{F(p)} = dF_p \circ d(F^{-1})_{F(p)} = d\text{Id}_N = \text{Id}_{T_{F(p)} N} .$$

Therefore, $(dF_p)^{-1} = d(F^{-1})_{F(p)}$ is a two-sided inverse of dF_p . Furthermore, dF_p is linear (i.e. a module (vector space) homomorphism) and thus

$$T_p M \cong T_{F(p)} N .$$

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